

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

Maven Dependencies :

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>1.7.30</version>
</dependency>

<dependency>
  <groupId>ch.qos.logback</groupId>
  <artifactId>logback-classic</artifactId>
  <version>1.2.3</version>
</dependency>
```

LoggingExample.java :

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args) {

        logger.error("This is an error message");

        logger.warn("This is a warning message");
    }
}
```

Output :

```
ERROR LoggingExample - This is an error message
WARN  LoggingExample - This is a warning message
```

Exercise 2: Parameterized Logging

Maven Dependencies :

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>1.7.30</version>
</dependency>

<dependency>
  <groupId>ch.qos.logback</groupId>
  <artifactId>logback-classic</artifactId>
  <version>1.2.3</version>
```

```
</dependency>
```

ParameterizedLoggingExample.java :

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class ParameterizedLoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(ParameterizedLoggingExample.class);

    public static void main(String[] args) {

        String name = "John";
        int age = 25;

        logger.info("User {} is {} years old", name, age);
    }
}
```

Output :

INFO ParameterizedLoggingExample - User John is 25 years old

Exercise 3: Using Different Appenders

Maven Dependencies :

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>1.7.30</version>
</dependency>

<dependency>
  <groupId>ch.qos.logback</groupId>
  <artifactId>logback-classic</artifactId>
  <version>1.2.3</version>
</dependency>
```

logback.xml :

```
<configuration>

  <appender name="console"
    class="ch.qos.logback.core.ConsoleAppender">
    <encoder>
      <pattern>
        %d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n
      </pattern>
    </encoder>
  </appender>

  <appender name="file"
    class="ch.qos.logback.core.FileAppender">
    <file>app.log</file>
    <encoder>
```

```

        <pattern>
            %d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n
        </pattern>
    </encoder>
</appender>

<root level="debug">
    <appender-ref ref="console"/>
    <appender-ref ref="file"/>
</root>

</configuration>

```

LoggingExample.java :

```

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args) {

        logger.info("Application started");

        logger.warn("This is a warning message");

        logger.error("This is an error message");
    }
}

```

Output :

```

10:30:15.123 [main] INFO  LoggingExample - Application started
10:30:15.125 [main] WARN  LoggingExample - This is a warning message
10:30:15.126 [main] ERROR LoggingExample - This is an error message

```